MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11738808 (BATTERY CHARGE SENSOR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Circuit Card Assembly: 11738808 (Battery Charge Sensor) (see 6. 1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manua1 Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedure for

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MIL-S-46944	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRA WINGS

U. S. Army, Frankfort Arsenal

F11738808	Circuit Card Assembly: 117388C3 (Battery Charge Sensor)

PACKAGING DATA SHEET

11738808	Packaging of Circuit Card Assembly: 11738808 (Battery Charge Sensor)

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing F11738808 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the following requirements, where applicable, of MlL-F-l 3926:

a. Order of precedence

b. Dimensions and tolerances

c. Part identification and marking

d. Electrical and electronic assemblies

e. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements of this specification at standard ambient texnpe ratures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Loads, poser, signals, and interconnections. The assembly shall perform as specified herein when the loads, power, signals, and interconnections of table I are applied as specified.

3.3.1.1 Battery power switching. With the power sources (item 2) of table I applied, and with an input voltage of +24.0 ±2.0 Vdc applied to P1-15, the voltage at Pi-5 shall be +24.O ±3.0 Vdc. With the input voltage at P1-15 decreased to +16.5 ±O.5 Vdc, the voltage at Pi-5 shall be +0.4 ±O.4 Vdc.

3.3.1.2 Malfunction (MALF) light disable. With the power source and input voltage of +24.O ±2.O Vdc applied as in 3.3.1.1, the output voltage at Pi-8 shall be +5.0 ±0.3 Vdc. With P1-22 and P1-7 disconnected, the output voltage at P1-8 shall be +0.2 ±0.2 Vdc. With the voltage at P1-15 disconnected, the output voltage at Pi-8 shall be +5.0 ±0.3 Vdc.

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TABLE I
LOADS, POWER, SIGNALS AND INTERCONNECTIONS

Item	Conditions	Characteristics	Connections
1.	Loads		Connected between the following pins:
1.1	Resistor	3.6 kilohm (kΩ) ±5% in series with +24 volts (V) (item 2.1)	P1-4(+) and P1-7(-)
1.2	Resistor	1.6 kΩ ±5 % in series with +24V (item 2.1)	P1-5(+) and P1-7(-)
1.3	Resistor	620Ω ±5%	P1-1 and P1-2
1.4	Resistor	5.6 kΩ ±5 %	P1-8 and P1-12
1.5	Resistor	620Ω ±5%	P1-11 and P1-2
1.6	Resistor	100Ω ±5%	P1-17 and P1-7
1.7	Resistor	100Ω ±5%	P1-18 and P1-7
2.	Power Source		
2.1	+24 Vdc	Tolerance: ±2.0V Ripple: ≤0.1 V p-p	See items 1.1 and 1.2
2.2	+15 Vdc	Tolerance: ±0.7 V Ripple: ≤0.025 V p-p	P1-2(+) and P1-7(-)
2.3	+5 Vdc	Tolerance: ±0.2 V Ripple: ≤0.025 V p-p	P1-12(+) and P1-7(-)
2.4	-6 Vdc	Tolerance: ±0.2 V Ripple: ≤0.025 V p-p	P1-10(-) and P1-7(+)
3.	Signal Source		
3.1	+17 to +24 Vdc		Applied as specified herein with return to P1-7(-)
4.	Interconnection		P1-7 and P1-22 except as specified herein.

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3.3.1.3 Battery drain light drive. With the power sources (item 2) of Table I applied and with the input voltage at P1-15 set to +24.0 ±2.0 Vdc, the voltage at P1-1 shall be 15.0 ±1.0 Vdc. With the voltage at P1-15 decreased to +16.5 ±0.5 Vdc, the voltage at P1-1 shall be + 1.0 ±0.7 Vdc.

3.3.1.4 Battery low light drive. With the power sources [item 2) of table I applied, and with +25.0 ±0.5 Vdc then +21.0 ±1.0 Vdc applied to pin P1-6, the output at pin P1-11 shall be +15.0 ±1.0 Vdc and +0.4 ±0.4 Vdc respectively.

3.3.1.5 MALF 1 signal. With the power sources (item 2) of table I applied, and with +24.0 ±0.2 Vdc applied to P1-6, the output at P1-14 shall be +0.2 ±0.2 Vdc. With P1-20 connected to P1-6, the output at P1-14 shall be +5.0 ±0.3 Vdc.

3.3.1.6 Converter drive signal. With the power sources (item 2) of table I applied, the output voltage at P1-17 and P1-18 shall be as shown in figure 1.

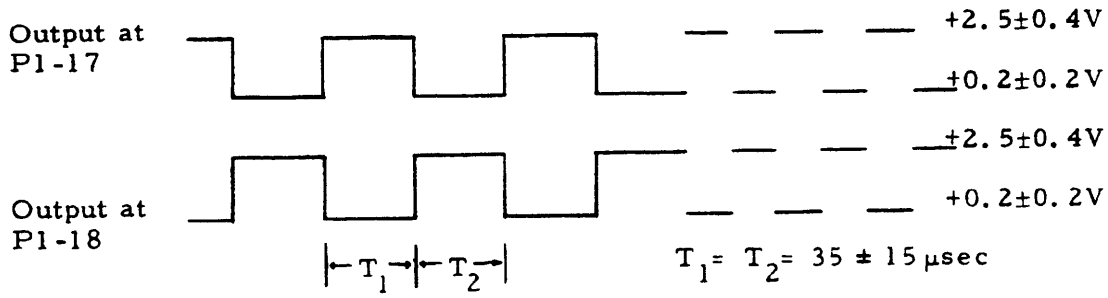


Figure 1. Converter Drive Signal Waveforms

3.3.2 Environmental

3.3.2.1 Storage temperature. The assembly shall meet the requirements of 3. 3.1 at ambient temperature after exposure to and thermal stabilizatio at -80°F and +160°F.

3.3.2.2 Operating temperatures. The assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3. 3.1 shall be met at ambient temperature.

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3.3.2.3 Vibration. The assembly shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a. Amplitude:	1/64 inch, constant (1/32 inch double amplitude)
b. Vibratory motion:	Simple harmonic
c. Sweep time:	5 to 55 to 5 Hertz in a 5 minute period
c. Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the conteract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.11 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for cc)nducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this insepction, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting the first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting o fficer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection or sampling procedures herein and submission of product provisions of MIL-STD-105.

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4. 3.2 Examination and tests

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Functional tests. The requirements and tests in table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

TABLE II
FUNCTIONAL TESTS

Classification	Characteristic	Requirement	Test Paragraph
101.	Battery power switching	3.3.1.1	4.6.2
102.	Malfunction (MALF) light disable	3.3.1.2	4.6.3
103.	Battery drain light drive	3.3.1.3	4.6.4
104.	Battery low light drive	3.3.1.4	4.6.5
105.	MALF 1 signal	3.3.1.5	4.6.6
106.	Converter drive signal	3.3.1.6	4.6.7

4.4 special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One assembly shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in table III.

TABLE III
SPECIAL SAMPLING - GENERAL

Classification	Control Test	Requirement	Test Paragraph
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

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4.4.2 Environmental. Three assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The cc]ntrol samples shall meet the requirements and tests in table IV.

TABLE IV
SPECIAL SAMPLING - ENVIRONMENTAL

Classification	Control Test	Requirement	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.8.1
304.	Operating temperature	3.3.2.2	4.6.8.1
305.	Vibration	3.3.2.3	4.6.8.2

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-1-45607. The inspection equipment listed in table V shall be utilized in performing the tests specified herein.

4.5.1 Government furnished inspection equipment. W'here the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment

4.5.2.2 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPS forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

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4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4. 6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine ca~ibration of inspection equipment.

TABLE V
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	DC power supply	+7.0 to +31.0 Vdc, adjustable Ripple: ≤ 100 mV p-p
2.	DC power supply	+24.0 ±2.0 Vdc Ripple: ≤100 mV p-p
3.	DC power supply	+5.0 ±0.1 Vdc Ripple: ≤25 mV p-p
4.	DC power supply	-6.0 ±0.1 Vdc Ripple: ≤25 mV p-p
5.	DC voltmeter	0 to 50 volts, accuracy of ±0.1%

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.1.1 Inition conditions. Apply the loads, interconnection, and power to the assembly as specified in table I.

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4.6.2 Batter power switching test. Apply the signal source specified in table I (item 3.1). Set the voltage at P1-15 to +24.0 +2.0 Vdc. Measure the voltage at P1-5 then set the voltage at P1-15 to +16.5 ±0.5 Vdc. Measure the voltage at P1-5. Verify compliance with 3.3.1.1.

4. 6.3 Malfunction (MALF) light disable test. Apply the signal source in table I (item 3.1) to P1-l5. Set the voltage to +24.0 ±2.0 Vdc. Measure the voltage at P1-8 to determine compliance with 3.3.1.2. Disconnect P1-22 and P1-7. Verify compliance with 3.3.1.2. Dis connect the voltage at P1-15. Verify compiance with 3.3.1.2.

4.6.4 Battery drain light drive test. Apply the signal source specified in table I (item 3. 1) to P1-15. Set the voltage P1-15 to +24.O ±2.O Vdc. Observe the voltage at P1-1 to determine compliance with 3.3.1.3. Decrease the voltage of P1-15 to +16.5 ±O.5 vdc. Verify compliance with 3.3.1.3.

4.6.5 Battery low light drive test. Apply the signal source specified in table I (item 3.1) to P1-6 and set, in turn, to +25.O ±O.5 Vdc and +21.0 ±1.0 Vdc. Measure the voltages at P1-11 to determine compliance with 3.3.1.4

4.6.6 MALF 1 signal test. Apply the signal source specified in table I (item 3. O) to Pi-6 and set to +24.O ±2.O Vdc. Measure the voltage at P1-14 to determine compliance with 3.3.1.5. Connect P1-20 to P1-6. Again measure the voltage at P1-14 to verify compliance with 3.3.1.5.

4.6.7 Coverter drive signal test. With the power sources (item 2) of table I applied, measure the outputs of P1-17, and P1-18 to determine compliance with 3.3.1.6.

4. 6.8 Environmental test.

4.6.8.1 Temperature test. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the assembly may be mounted on the inside surface of the test chamber. Perform this test in accordance with procedure I of MIL-F-13926, with conditions C (-40°F), and E (+125°), and with the rate of

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temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4. 6.2 through 4. 6.7 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.8.2 Vibration test. Perform this test utilizing the apparatus specified in method 514.1 of MIL-STD-810 and in accordance with part I of procedure I therein, except that the vibratory conditions of 3.3. 2. 3 shall apply. Following vibration exposure, examine the assembly for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.2 through 4.6.7 to determine compliance with 3.3.2.3.

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a. To provide converter-drive signal for the battery charging circuit

b. To provide a signal indicating a malfunction in the battery power supply

c. To provide a signal indicating a low battery voltage condition

d. To energize the battery switch when the prime vehicle power level falls below 18 Vdc.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable Levels of preservation, packaging, and packing

d. Applicable stock number

e. Requirement for First Article Submission.

Custodian:

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Preparing Activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE: MIL-C-50740(MU) Circuit Card Assembly: 11738808 (Battery Charge Sensor)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

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